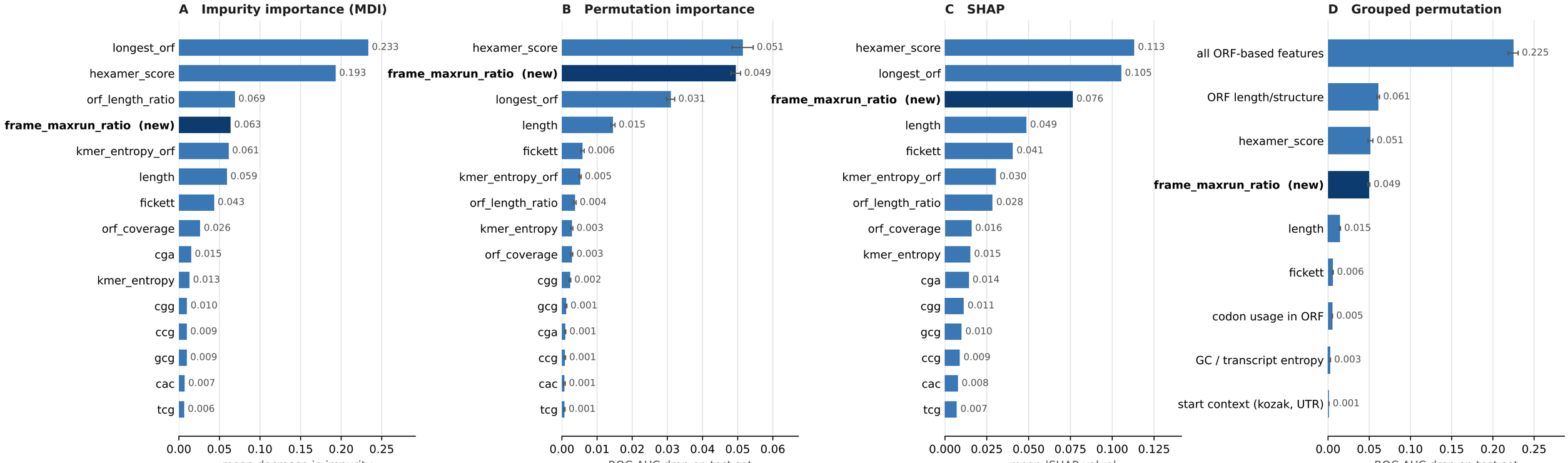

RAIDNC v2 (compact). MDI from training; permutation importance on the independent test set (3,000 + 3,000; baseline ROC-AUC 0.973; mean ± SD of 10 permutations); SHAP on 2,000 balanced test sequences.
D: correlated features permuted jointly. "All ORF-based features" = ORF length/structure + hexamer + codon usage + Kozak/UTR.